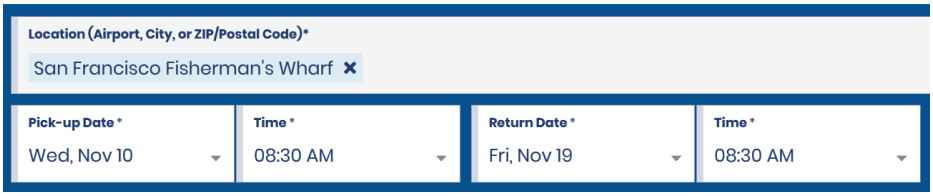
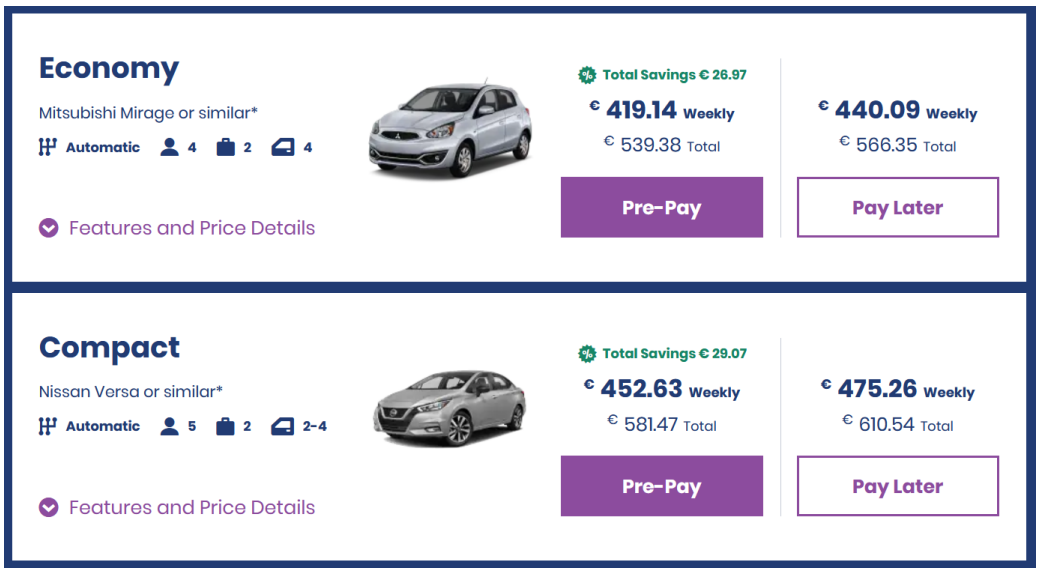
Assignment #1

Online Rent a Car

The aim is to represent the management of a car rental company. The typical rental process is the following: first, customers book via a web portal a car, then pick up the car, drive and finally return it.
In a first step, the customer starts a reservation by mentioning where they want to pick up the car and the desired agenda



Next, the customer chooses which kind of car she wants. Each possible category is described with the help of a name, typical number of passengers and windows, gearbox, and also a typical car.



Notice that the pre-pay price is lower than the pay later-one: paying later means paying when picking the car at the rental agency. However, pre-pay involves being charged if the reservation is cancelled.

Some insurances like damage Waiver are included but the customer can add extra insurances (described via a name and a price per day):



The customer may also choose extra options (baby seat, additional driver…). Each of these options has a name, price, and picture. Once her reservation is fully described, the customer needs to sign in to book. Once connected, the customer will provide her details (email, first and last name, address which includes street, city, country, and zip code); if the customer chooses the prepay option, she then pays by credit card.

Later on, the customer may cancel her reservation, if she previously booked with the pre-pay option, she will have to pay a cancellation fee. To do so, she connects to the system, sign in, and request to cancel her reservation.

On the day of the reservation, the customer will go the location to pick up a car. Based on the chosen category, the agency employee will allocate a specific car to the customer. The allocated car has a brand, a colour, and a licence plate number and of course a category. Before giving the keys to the customer, the employee will first check and scan the driving licence of the customer. Next, the employee inspects the car together with the customer and ask him to sign the evaluation of the car (no damage for instance). Signature is digital.

After the customer uses the car, she will return to the pick in location: employee checks the condition of the car and ask to the customer to sign the evaluation. Notice that the employee records mileage and date and time of the return operation.

If the customer has a problem (accident or breakdown), she can use the mobile app to contact the rental company. She must provide all possible details: current location, time and date of the problem and circumstances; she can also upload photos. The 24/7 hotline will contact her back and organize with her the repairing actions.

To do

Create the use case diagram corresponding to the overall rental process

Detail the use cases associated to the action “pick up car” with the help of a sequence system diagram (you may provide additional lo-fi interfaces)

Create a high-level class diagram (main classes and key properties)

Detail via a sequence diagram the interaction between objects associated either to a “reservation action” or a “cancel action”

Deliverables and agenda

Write down a report containing your diagrams and all information explaining your work. Notice that all diagrams should be commented: explain you design choices. Group work is allowed (max size of the group: 2). Your report should be uploaded by October 30, 23:55. An assessment will hold on October 30